GANs (Theory) and Autoencoders on FashionMNIST

Ritika Mukesh Neema | DATA 266 | Step 0: SID4=6638, SEED=6638, SLICE=638, HP_ID=2, CLS_A=8, CLS_B=5 | Colab, Tesla T4

As permitted, the second hyperparameter configuration is skipped, since this assignment already trains two different models (FC-AE and VAE). All results below come from the executed notebook.

Part 1 — GAN Concepts

1.1 Architecture, roles and the adversarial objective

A GAN (Goodfellow et al., 2014) trains two networks against each other. The **generator** G maps a noise vector z (Gaussian or uniform) to a synthetic sample G(z); it never sees real data and learns only from the gradient that flows back through the discriminator. The **discriminator** D is a binary classifier that outputs the probability that an input is real rather than generated. They optimize the minimax game

$$\min_G \max_D V(D,G) = E_{x\sim p_data}[\log D(x)] + E_{z\sim p_z}[\log(1 - D(G(z)))]$$

Interaction. Training alternates. (1) *Discriminator step:* with G frozen, D is updated on a real batch (label 1) and a fake batch (label 0) using binary cross-entropy. (2) *Generator step:* with fresh noise, G is updated so that D labels its outputs as real. In practice the non-saturating loss max log D(G(z)) replaces min log(1 − D(G(z))) because the latter gives vanishing gradients early on, when D easily rejects poor fakes. D therefore acts as a learned, adaptive loss: as it finds new flaws, G is pushed to remove them.

Theoretical goal. For fixed G the optimal discriminator is D*(x) = p_data(x) / (p_data(x) + p_g(x)). Substituting it back, the generator minimizes –log 4 + 2·JSD(p_data || p_g), which reaches its minimum if and only if p_g = p_data. At that equilibrium D*(x) = 1/2 everywhere, so the discriminator is no better than chance. The guarantee assumes infinite capacity and an optimal D at each step; with finite networks and simultaneous gradient updates, training can oscillate or fail, which motivates the techniques in 1.2.

1.2 Mode collapse

Definition. Mode collapse is when G produces only a small subset of the modes of the real distribution (in the extreme, a single output) regardless of z. Samples may look sharp but diversity is lost: p_g covers only part of the support of p_data. On FashionMNIST this would look like a generator that draws only sneakers.

Why it occurs.

- G is rewarded for fooling the *current* discriminator, not for covering all of p_data. If one mode is rated most “real”, mapping every z to it is the easiest way to lower the loss.
- With alternating updates, G effectively optimizes against a snapshot of D rather than the true min-max order. G jumps to the region D currently likes, D learns to reject it, and G hops to another mode, cycling instead of covering the distribution.
- D scores each sample independently, so nothing penalizes repeating one good sample many times.
- When D becomes too strong, gradients vanish or become uninformative, removing the signal that would push G toward under-covered regions.

Mitigation techniques.

- **Minibatch discrimination (Salimans et al., 2016).** D receives features that compare each sample with the others in the same batch (a learned similarity, $\exp(-\|M_i - M_j\|_{L1})$). If G outputs near-identical samples, the within-batch similarity is abnormally high and D can flag the batch as fake, so G can no longer lower its loss by repeating one mode and is forced to produce diverse batches.

- **Wasserstein GAN with gradient penalty (Arjovsky et al., 2017; Gulrajani et al., 2017).** The JS-based loss is replaced by the Earth-Mover distance, estimated by a critic constrained to be 1-Lipschitz (a penalty $\lambda \cdot E[(||\text{grad } f(x_hat)||_2 - 1)^2]$). Unlike JSD, this distance gives a smooth, non-saturating gradient even when p_g and p_data have disjoint support, so the critic can be trained to optimality without starving G of gradient. G keeps receiving a signal to move mass toward modes it has missed, and the critic loss correlates with sample quality.
- **Unrolled GANs (Metz et al., 2017).** G's update is computed against D after k look-ahead update steps of D (backpropagating through them). G sees how D will react, so it cannot exploit a momentary weakness by collapsing onto one mode that D would reject a few steps later.

1.3 Three significant GAN variants

DCGAN (Radford et al., 2016)

- **Problem:** the original MLP-based GAN was unstable and scaled poorly to images.
- **Innovations:** an all-convolutional design with guidelines: strided convolutions in D and transposed convolutions in G replace pooling, so both learn their own resampling; batch normalization in both networks; no fully connected hidden layers; ReLU in G (tanh at the output) and LeakyReLU in D. It also showed that z has semantic structure (vector arithmetic on z gives meaningful edits).

Conditional GAN (Mirza & Osindero, 2014)

- **Problem:** a vanilla GAN gives no control over what is generated, since z is unstructured.
- **Innovations:** both networks receive extra conditioning information y (a class label, text or another image); G(z|y) and D(x|y) concatenate y with their inputs. D must judge not only realism but whether the sample matches the condition. This enables class-conditional generation and underlies pix2pix and text-to-image GANs; conditioning also partly discourages mode collapse because each condition must be covered.

CycleGAN (Zhu et al., 2017)

- **Problem:** image-to-image translation (horse↔zebra, photo↔painting) normally needs paired training images, which are often impossible to collect.
- **Innovations:** two generators G: X→Y and F: Y→X with two discriminators, trained on unpaired collections. The key addition is the **cycle-consistency loss** $||F(G(x)) - x||_1 + ||G(F(y)) - y||_1$, which forces translations to preserve the input's content (otherwise G could map every x to any realistic y). An optional identity loss helps preserve colour.

Part 2 — FC-AE vs. VAE on FashionMNIST

2.1 Setup

Both models use the same settings so the comparison is fair: 60,000 training and 10,000 test images with pixels in [0, 1]; latent size 32; hidden layers 784→512→256→32 (decoder mirrors this, sigmoid output); Adam, learning rate 1e-3; batch size 128; 20 epochs; seed 6638. The FC-AE minimizes per-pixel MSE. The VAE uses a Gaussian encoder (mean μ, log-variance) with the reparameterization trick and minimizes binary cross-entropy + KL(q(z|x) || N(0, I)), both per image.

2.2 Quantitative results (test set)

Metric	FC-AE	VAE
Parameters	1,083,696	1,091,920
Final train loss	0.00818 (MSE/px)	238.19 (-ELBO/img)
Final test loss	0.00843 (MSE/px)	240.33 (-ELBO/img)
Test reconstruction MSE / pixel	0.00843	0.01510
Test PSNR (dB)	20.74	18.21
VAE test BCE / KL per image	-	228.39 / 11.93
Linear-probe test accuracy (32-d latent)	0.8267	0.8068
Training time (s, T4)	190.5	196.5

The VAE's loss is the negative ELBO per image, so its train/test loss is not comparable with the FC-AE's MSE; the reconstruction-MSE and PSNR rows are the like-for-like comparison. The linear probe fits a logistic regression on the frozen 32-d latents (the VAE uses the posterior mean).

2.3 Training curves

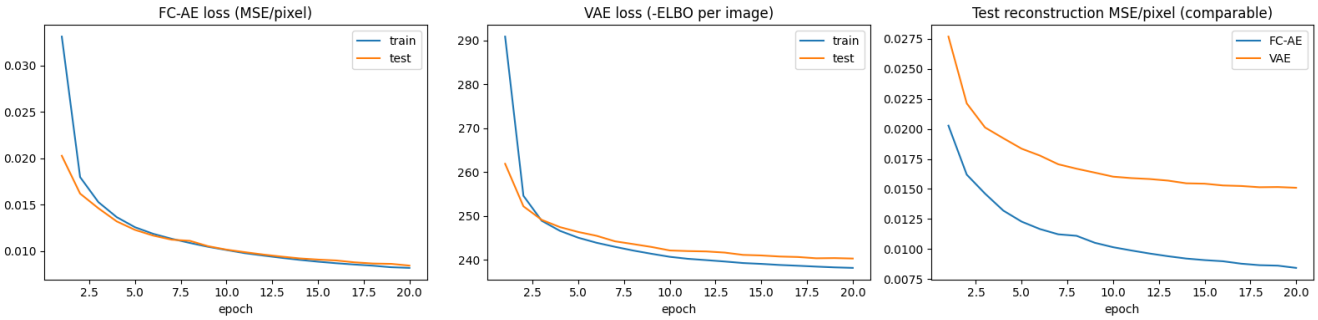


Figure 1. Left: FC-AE loss. Middle: VAE loss (negative ELBO per image). Right: test reconstruction MSE per pixel for both models, which is directly comparable.

2.4 Reconstructions

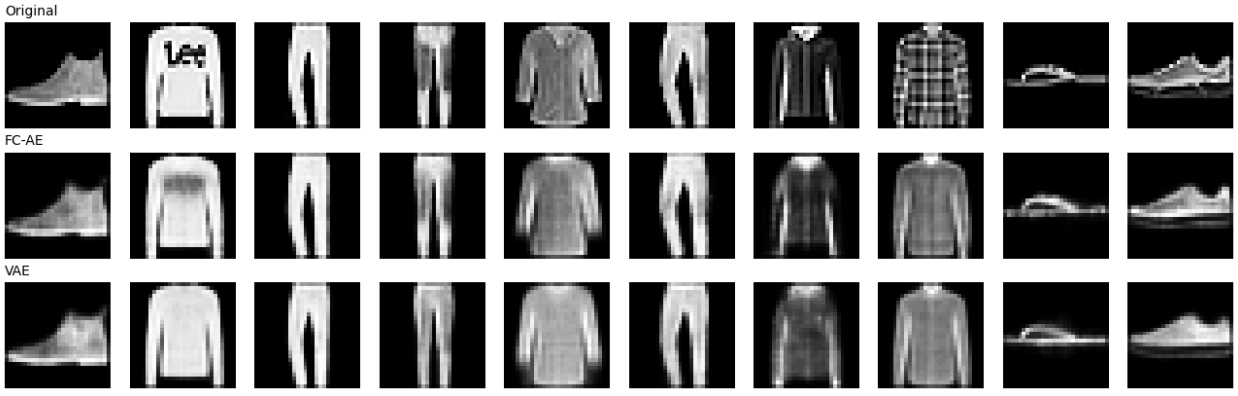


Figure 2. The same ten test images: originals, FC-AE reconstructions, VAE reconstructions.

2.5 Generation from the prior

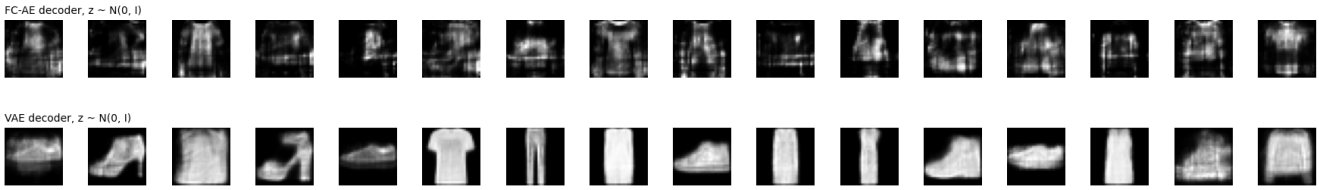


Figure 3. Decoding z ~ N(0, I) with each model's decoder.

2.6 Latent space (t-SNE of 3,000 test latents)

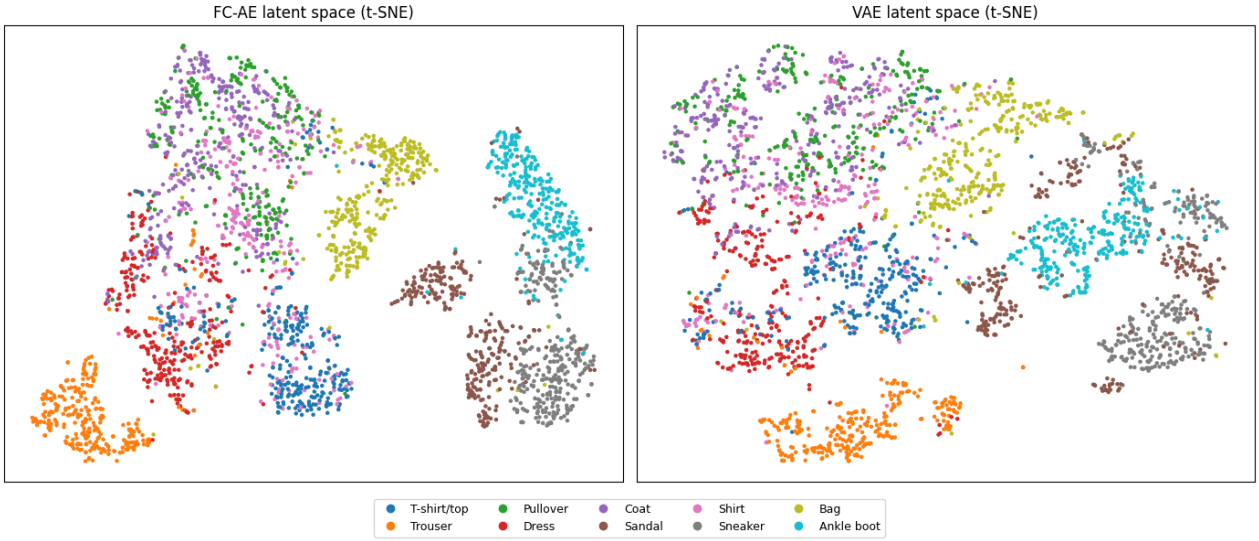


Figure 4. t-SNE of the 32-d latent codes, coloured by class.

2.7 Latent interpolation

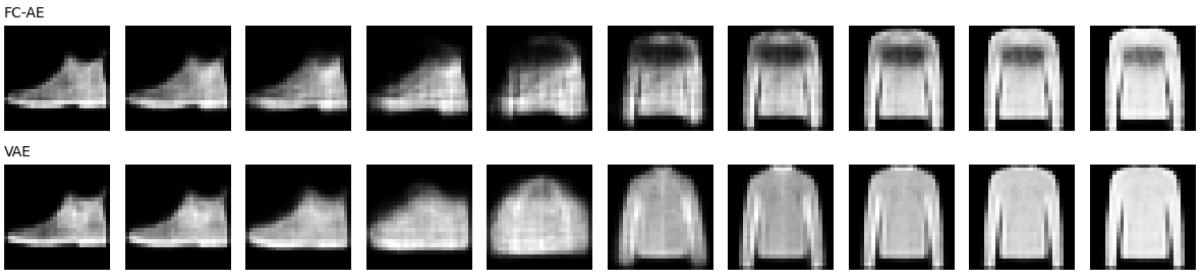


Figure 5. Linear interpolation between the codes of an ankle boot and a pullover.

2.8 Analysis

Reconstruction: the FC-AE wins. Its test MSE is 0.00843 per pixel (20.74 dB) versus 0.01510 (18.21 dB) for the VAE, roughly 44% lower error. The FC-AE keeps finer detail (the plaid shirt, jacket stripes, the sneaker sole); the VAE outputs are smoother. This is a trade-off, not a flaw: the FC-AE optimizes reconstruction only, while the VAE also pays a KL cost (about 12 nats per image against about 228 for reconstruction). The KL term makes the encoder's posteriors noisy and overlapping, so the decoder cannot rely on exact latent coordinates and outputs the average of everything compatible with a code, which is the blur. The FC-AE test MSE was still falling in the last epochs (0.00866, 0.00862, 0.00843), so longer training would widen the gap slightly.

Generation: only the VAE works. Decoding z ~ N(0, I) gives recognizable garments from the VAE (sneakers, ankle boots, a high-heeled shoe, a T-shirt, trousers, dresses), though some are blurry, because the KL term pulls

every posterior toward the standard normal so the region the decoder was trained on matches the region we sample. The FC-AE decoder gives noisy, fragmented shapes from the same noise: nothing constrains its codes to any distribution, so random vectors land in “holes” between the codes it has seen. The FC-AE is a good compressor but not a generative model.

Latent structure. Both t-SNE maps separate the visually distinct classes (trousers, bags, ankle boots, footwear) and both confuse the same hard group (shirt, T-shirt, pullover, coat), as expected for an unsupervised 32-d code. The linear probe agrees: 82.7% (FC-AE) vs. 80.7% (VAE), so the VAE's regularization costs little class information. The two maps look broadly similar, with the VAE's clusters somewhat more diffuse and adjacent.

Interpolation. The FC-AE interpolation stays on the boot for several steps and then flips abruptly through a degraded, texture-heavy intermediate, indicating gaps in its latent space. The VAE transitions more gradually, though its midpoint is a blurry blob rather than a crisp object, which is the practical payoff of KL regularization: a smoother, more continuous latent space.

Takeaway. Use the FC-AE for faithful reconstruction or compact features (compression, denoising, feature extraction); use the VAE for generation or a smooth, samplable latent space. Possible improvements: convolutional encoders/decoders for sharper images from both models; a larger latent size or a KL weight $\beta < 1$ for the VAE to trade smoothness for detail; and a perceptual or adversarial loss (e.g. VAE-GAN, linking back to Part 1) to reduce VAE blur.